

off-policy Evaluation

policy of interest

The goal is to evaluate a target policy using samples from another policy, called a behavior policy.

policy generating behavior

π^t : target policy

π^b : behavior policy

Benefits

Enables using samples from

- old policies
- exploratory policies
- other agents / demonstrations

Note: Consider a sample finite trajectory from an MDP

rewards conform
to the deterministic
reward function

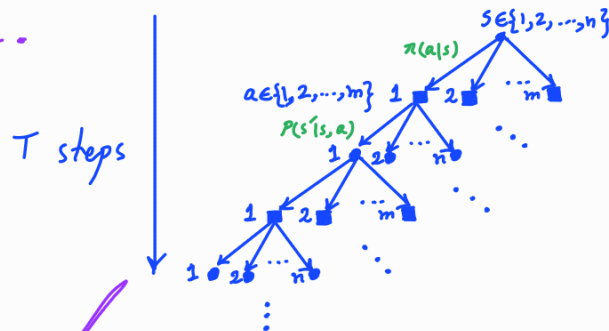
$$z_T = (s_0, a_0, r_0, s_1, a_1, r_1, \dots, s_{T-1}, a_{T-1}, r_{T-1}, s_T)$$

$$r_0 = R(s_0, a_0)$$

$$r_1 = R(s_1, a_1)$$

$$r_{T-1} = R(s_{T-1}, a_{T-1})$$

generated based on an initial distribution μ_0 and following a stationary, stochastic policy π .



* The probability of z_T being realized is

$$P(z_T | \mu_0, \pi, P) = P(\overbrace{s_0, a_0, s_1, a_1, \dots, s_{T-1}, a_{T-1}, s_T}^{z_T} | \mu_0, \pi, P)$$

$$= P(s_T | s_0, a_0, s_1, a_1, \dots, s_{T-1}, a_{T-1}, \mu_0, \pi, P)$$

$$P(s_T | s_{T-1}, a_{T-1})$$

$$P(a_{T-1} | s_0, a_0, s_1, a_1, \dots, s_{T-1}, \mu_0, \pi, P)$$

$$\pi(a_{T-1} | s_{T-1})$$

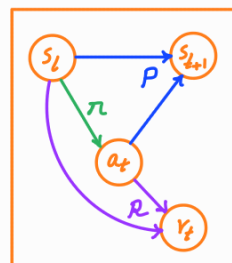
$$P(\overbrace{s_0, a_0, s_1, a_1, \dots, s_{T-1}}^{z_{T-1}} | \mu_0, \pi, P)$$

$$=$$

repeat the factorization

$$\vdots$$

$$= \mu_0(s_0) \prod_{t=0}^{T-1} \pi(a_t | s_t) P(s_{t+1} | s_t, a_t)$$



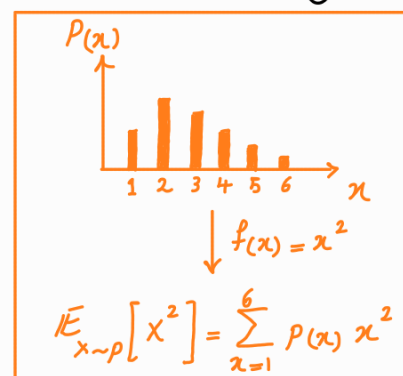
$z_T \rightarrow z_{T-1} \rightarrow z_{T-2} \rightarrow \dots \rightarrow z_1 \rightarrow z_0$
unroll trajectory over time

Importance sampling: A general technique for evaluating properties of a target distribution, e.g., the expectation of a function over that distribution based on samples from another distribution.

Let X be a discrete RV distributed according to probability measure P .

$$\mathbb{E}_{X \sim P} [f(X)] = \sum_{x \in X} P(x) f(x)$$

$$\mathbb{E}_{X \sim P} [X] = \sum_{x \in X} P(x) x$$



Having n i.i.d. samples $x_1, x_2, \dots, x_n$ of X , drawn from $P(X)$, a simple Monte Carlo method can estimate this expected value by the empirical mean

$$\hat{\mathbb{E}}_{X \sim P} [f(X)] = \frac{1}{n} \sum_{i=1}^n f(x_i)$$

Now, consider another distribution over X captured by probability measure Q .

$$\begin{aligned} \mathbb{E}_{X \sim P} [f(X)] &= \sum_{x \in X} P(x) f(x) = \sum_{x \in X} Q(x) \underbrace{\frac{P(x)}{Q(x)}}_{\text{Likelihood ratio weight } w(x)} f(x) \\ &= \mathbb{E}_{X \sim Q} \left[\underbrace{\frac{P(x)}{Q(x)} f(x)}_{g(x)} \right] \end{aligned}$$

requirement:
 $Q(x) > 0$ if $P(x) f(x) \neq 0$

Having n i.i.d. samples $x_1, x_2, \dots, x_n$ of X , drawn from $Q(X)$, a simple Monte Carlo method can estimate this expected value by the empirical mean

$$\hat{E}_{x \sim p} [f(x)] = \hat{E}_{x \sim Q} \left[\frac{P(x)}{Q(x)} f(x) \right] = \frac{1}{n} \sum_{i=1}^n \frac{P(x_i)}{Q(x_i)} f(x_i)$$

$w(x_i)$ should be known

Importance sampling for off-policy evaluation: Let G_T be the sample return corresponding to sample trajectory z_T generated by π^b .

$$w(z_T) = \frac{P(z_T | \mathcal{M}_0, \pi^t, P)}{P(z_T | \mathcal{M}_0, \pi^b, P)} = \frac{\mathcal{M}_0(s_0) \prod_{t=0}^{T-1} \pi^t(a_t | s_t) P(s_{t+1} | s_t, a_t)}{\mathcal{M}_0(s_0) \prod_{t=0}^{T-1} \pi^b(a_t | s_t) P(s_{t+1} | s_t, a_t)}$$

P is unknown but it cancels out

$$= \prod_{t=0}^{T-1} \frac{\pi^t(a_t | s_t)}{\pi^b(a_t | s_t)}$$

$$z = (s_0=1, a_0=2, s_1=3, a_1=0, s_2=1, a_2=1)$$

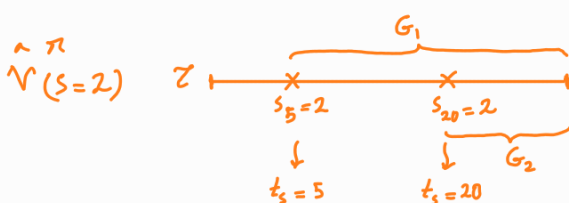
$$w(z) = \frac{\pi^t(2|1)}{\pi^b(2|1)} \times \frac{\pi^t(0|3)}{\pi^b(0|3)} \times \frac{\pi^t(1|1)}{\pi^b(1|1)}$$

off-policy MC evaluation:

the return of the trajectory

$$E_{\pi^t, P} [G_T] \approx \hat{E}_{\pi^b, P} \left[\prod_{t=0}^{T-1} \frac{\pi^t(a_t | s_t)}{\pi^b(a_t | s_t)} G_T \right] = \frac{1}{n} \sum_{i=1}^n w_i G_i$$

$$\hat{V}^{\pi^t}(s) \leftarrow \hat{V}^{\pi^t}(s) + \alpha \left(\prod_{t=0}^{T-1} \frac{\pi^t(a_t | s_t)}{\pi^b(a_t | s_t)} G_T - \hat{V}^{\pi^t}(s) \right)$$



$t=t_s$ if the trajectory does not start from s

off-policy TD evaluation:

$$\mathbb{E}_{\pi^t, p} [R(s_t, A_t) + \gamma V(s_{t+1}) \mid s_t = s]$$

$$= \mathbb{E}_{\substack{A_t \sim \pi^t(s) \\ s_{t+1} \sim p(\cdot | s_t, A_t)}} [R(s_t, A_t) + \gamma V(s_{t+1}) \mid s_t = s]$$

$$\approx \mathbb{E}_{\substack{A_t \sim \pi^b(s) \\ s_{t+1} \sim p(\cdot | s_t, A_t)}} \left[\frac{\pi^t(A_t | s_t)}{\pi^b(A_t | s_t)} (R(s_t, A_t) + \gamma V(s_{t+1})) \mid s_t = s \right]$$

$$\hat{V}^{\pi^t}(s) \leftarrow \hat{V}^{\pi^t}(s) + \alpha \left(\frac{\pi^t(A_t | s_t)}{\pi^b(A_t | s_t)} (R(s_t, A_t) + \gamma \hat{V}^{\pi^t}(s_{t+1})) - \hat{V}^{\pi^t}(s_t) \right)$$

$$\text{new value} \leftarrow \text{previous value} + \text{learning rate} \underbrace{(\text{target} - \text{previous value})}_{\text{weighted target}}$$

off-policy Optimization

The goal is to optimize a target policy using samples from another policy, called a behavior policy.

We can use a generalized policy iteration approach, where the policy evaluation step is performed using importance sampling.

off-policy MC method with soft exploration

Iterative method:

- Initialize $\hat{Q}$ $\pi^t(s) = \arg\max_{a \in A} \hat{Q}(s, a)$
- Initialize π^t as a greedy policy with respect to $\hat{Q}$
- Initialize $\mathcal{G}(s, a) = \emptyset$ for all $s \in S, a \in A$
- Repeat
 - Build π^b as a soft policy with respect to $\hat{Q}$
 - Generate a sample trajectory τ using π^b

policy
evaluation

- For $(s, a) \in S \times A$ s.t. $(s, a) \in \tau$:
 - $t_{s,a} =$ first time (s, a) is visited
 - $G = \prod_{t=t_{s,a}}^T \frac{\pi^t(a_t | s_t)}{\pi^b(a_t | s_t)} \left(\sum_{t=t_{s,a}}^T \gamma^{t-t_{s,a}} R(s_t, a_t) \right)$
 - $\mathcal{G}(s, a) \leftarrow \mathcal{G}(s, a) \cup \{G\}$
 - $N(s, a) \leftarrow N(s, a) + 1$
 - $\hat{Q}(s, a) = \frac{1}{N(s, a)} \sum_{G \in \mathcal{G}(s, a)} G$

policy
improvement

- Build π^t as a greedy policy with respect to $\hat{Q}$

off-policy TD method (Q learning)

Q learning does not use importance sampling

Iterative method:

- Initialize $\hat{Q}$ s.t. $\hat{Q}(s,a)=0$ for terminal states
- Initialize π^t as a greedy policy with respect to $\hat{Q}$
- Repeat
 - Sample $s \in S$ $s_0 \sim \mathcal{M}_0$
 - Repeat until the episode terminates
 - Build π^b as a soft policy with respect to $\hat{Q}$ (e.g., ϵ -greedy)
 - Take action a according to $\pi^b(a|s)$ at s
 - Observe reward $R(s,a)$ and next state s'

policy evaluation and improvement

$$\hat{Q}(s,a) \leftarrow \hat{Q}(s,a) + \alpha [R(s,a) + \gamma \max_{a' \in A} \hat{Q}(s',a') - \hat{Q}(s,a)]$$

policy improvement

- Build π^t as a greedy policy with respect to $\hat{Q}$
 - Update the current state $s \leftarrow s'$

Note: While SARSA uses Bellman consistency equation to update the Q function, Q learning uses the Bellman optimality equation.

- Bellman consistency equation for $Q^\pi(s, a)$:

$$Q^\pi(s, a) = R(s, a) + \gamma \mathbb{E}_{\substack{s' \sim p(\cdot | s, a) \\ a' \sim \pi(\cdot | s)}} [Q^\pi(s', a')]$$

- Bellman optimality equation for $Q^*(s, a)$:

$$Q^*(s, a) = R(s, a) + \gamma \mathbb{E}_{s' \sim p(\cdot | s, a)} \left[\max_{a' \in \mathcal{A}} Q^*(s', a') \right]$$